

Lab Task 4: Morphological Analysis for Word Embedding

Aim:

To develop algorithms for finding the morphology of word documents using stemming, lemmatization, and morphological features with NLTK and spaCy in Python.

CODE:

```
# Install (run once)

# !pip install nltk spacy

import nltk

import spacy

from nltk.tokenize import word_tokenize

from nltk.stem import PorterStemmer

# Download required data

nltk.download('punkt')

nltk.download('punkt_tab')

# Load spaCy model

# python -m spacy download en_core_web_sm

nlp = spacy.load("en_core_web_sm")

# Sample text

text = "The boys are playing games and running faster than before."

# ♠ Tokenization

tokens = word_tokenize(text.lower())

# ♠ Stemming (NLTK)

stemmer = PorterStemmer()

stemmed_words = [stemmer.stem(word) for word in tokens]

print("NLTK Stemming Output:")

for word, stem in zip(tokens, stemmed_words):

    print(word, "->", stem)

# ♠ spaCy Morphological Analysis

doc = nlp(text)

print("\nspacey Morphological Analysis:")
```

```
for token in doc:

    print("Word:", token.text)

    print(" Lemma:", token.lemma_)

    print(" POS:", token.pos_)

    print(" Morphology:", token.morph)

    print()
```

output:

NLTK Stemming Output:

boys -> boy
playing -> play
games -> game
running -> run
before -> befor

spaCy Morphological Analysis:

Word: boys
Lemma: boy
POS: NOUN
Morphology: Number=Plur
Word: playing
Lemma: play
POS: VERB
Morphology: Tense=Pres|VerbForm=Part
Word: running
Lemma: run
POS: VERB
Morphology: Tense=Pres|VerbForm=Part
Word: faster
Lemma: fast
POS: ADV
Morphology: Degree=Cmp